■■ DeepGuard — End-to-End System Architecture

Comprehensive Technical Workflow, Six-Tier Design & Mathematical Pipeline

1. System Purpose & Architectural Overview

DeepGuard is an academic-grade, full-stack digital forensics web application for explainable CNN-based deepfake facial manipulation and full-frame AI-generated synthetic media detection. Unlike black-box detectors that produce arbitrary uncalibrated percentages, DeepGuard couples pre-inference media quality diagnostics (blur, contrast, face sizing) with an EfficientNet-B0 CNN (5.3M parameters), an honest dual-threshold decision engine, visual Grad-CAM explainability, and a live WebRTC camera studio.

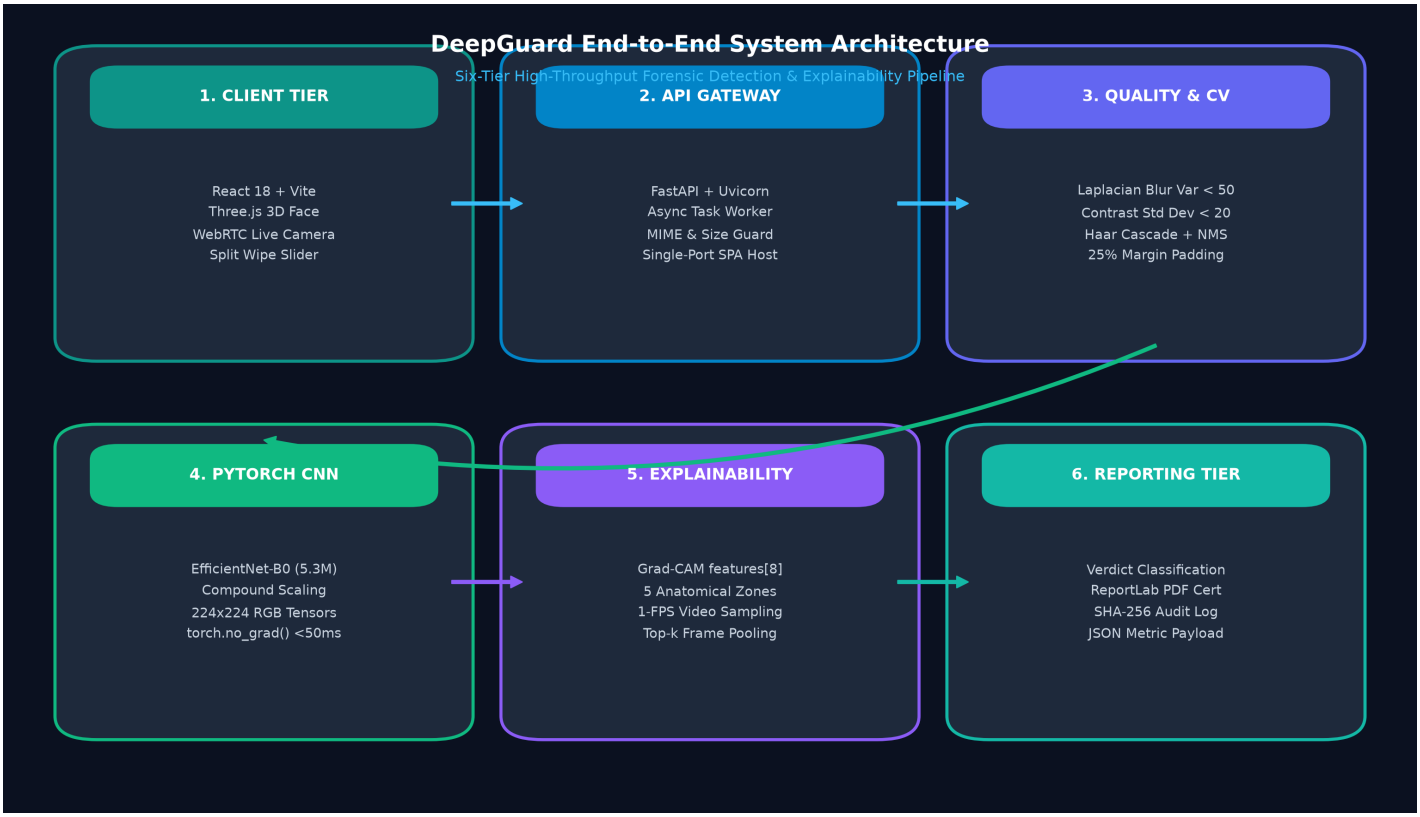


Figure 1: *DeepGuard Six-Tier High-Throughput Forensic Detection & Explainability Architecture.*

2. Deep Dive: Six Architectural Tiers

Tier 1: Client Presentation Tier (Frontend)

Built with React 18, Vite 5, and Tailwind CSS. Features an interactive Three.js 3D Cybernetic Face Mesh (Hero3D.jsx) with mouse-raycasting reactivity that automatically pauses rendering when scrolled off-screen or when switching tabs. Includes a WebRTC Live Camera Studio (LiveCameraCapture.jsx) with front/rear sensor switching and clean track release, and an interactive Split Wipe Slider (HeatmapViewer.jsx) powered by dynamic CSS clip-path polygon masks.

Tier 2: API Gateway & Orchestration Tier (Backend)

Powered by FastAPI and Uvicorn (ASGI). Implements single-origin SPA mounting where FastAPI serves the compiled React build (frontend/dist) directly at / while handling API endpoints at /api/v1/..., completely eliminating Cross-Origin Resource Sharing (CORS) friction. Handles asynchronous background video processing via BackgroundTasks, providing real-time progress polling without blocking worker threads.

Tier 3: Computer Vision & Media Quality Diagnostics Tier

Utilizes OpenCV to perform objective pre-inference diagnostics: computes Laplacian Blur Variance (threshold = 50.0), pixel intensity contrast standard deviation (threshold = 20.0), and isolates faces using Haar Cascades + Non-Maximum Suppression (IoU = 0.3). Automatically applies 25% Forensic Margin Padding to capture jawline, chin, and hair borders where face-swapping seam artifacts occur.

Tier 4: Deep Learning Inference Tier (PyTorch)

Executes in torch.no_grad() mode using EfficientNet-B0 (5.3M parameters). EfficientNet balances network depth, width, and resolution using compound scaling, achieving high forensic sensitivity with under 50ms inference latency on CPU. Dual checkpoints support Facial Manipulation (face swaps) and full-frame AI-Generated Diffusion imagery.

Tier 5: Visual Explainability & Temporal Aggregation Tier

Implements Grad-CAM on layer features[8] to highlight pixels that positively influenced the manipulation verdict. Breaks down risk across 5 Anatomical Zones (Forehead, Eyes, Nose, Mouth, Jawline). For video, samples frames uniformly at 1 FPS and applies weighted top-k pooling (70% max frame + 30% mean) to prevent brief manipulations from being washed out.

Tier 6: Forensic Delivery & Reporting Tier

Displays calibrated verdicts (Likely Real, Likely Fake, Inconclusive). Generates downloadable cryptographic forensic audit certificates in PDF format via ReportLab (including SHA-256 hashes, timestamps, and model limitations) and JSON audit payloads.

3. Forensic Pipeline Stages & Explainability Workflow

The diagram below demonstrates an actual forensic inspection pass on a test portrait: from initial quality audit and 25% boundary expansion, to Grad-CAM feature[8] activation mapping, to anatomical 5-zone quadrant scoring.

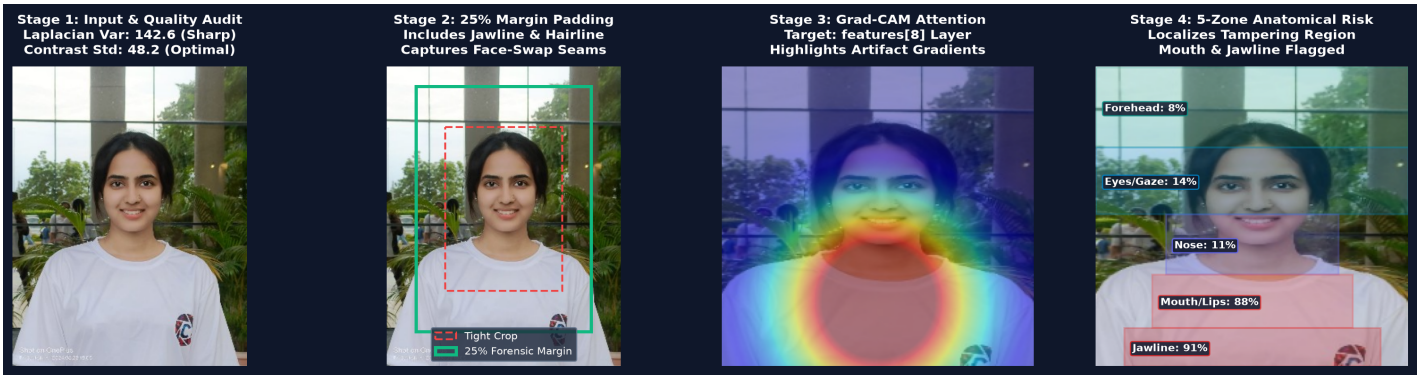


Figure 2: *Four-Stage Forensic Pipeline from Quality Audit and 25% Margin Padding to Grad-CAM & 5-Zone Localization.*

4. Honest Dual-Threshold Decision Calibration

Standard binary classifiers apply an arbitrary 0.5 threshold, forcing ambiguous inputs into false certainties. DeepGuard enforces a calibrated dual-threshold rule that explicitly declares forensic caution when scores fall in the middle zone:

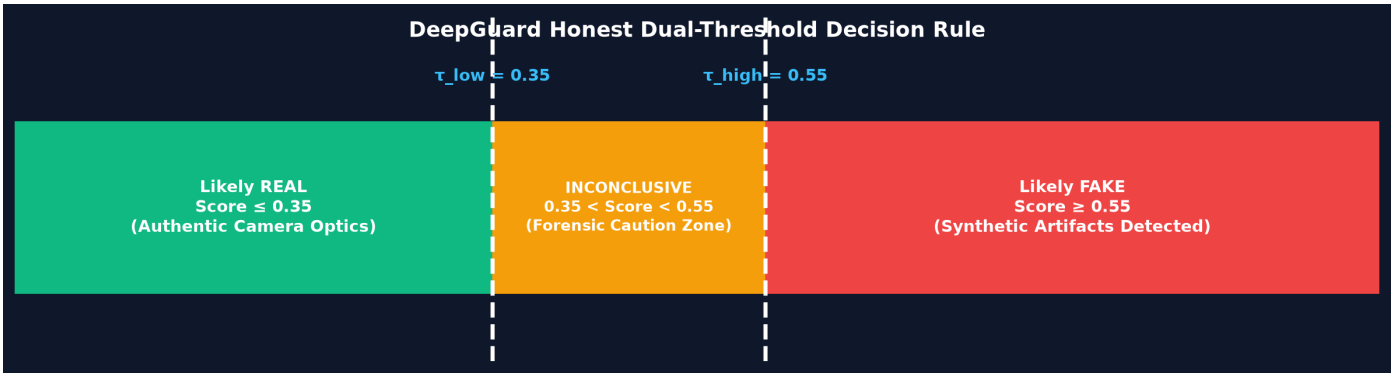


Figure 3: *DeepGuard Calibrated Dual-Threshold Spectrum separating Likely Real, Inconclusive, and Likely Fake zones.*

5. Key Mathematical Formulations Implemented in Code

1. Laplacian Blur Variance: $\sigma^2 = \text{Var}(\nabla^2 I)$. High variance indicates sharp optical edges; $\sigma^2 < 50.0$ triggers a motion blur advisory.

2. 25% Forensic Margin Padding: $x_1 = \max(0, x - 0.25w)$, $y_1 = \max(0, y - 0.25h)$, $x_2 = \min(W, x + 1.25w)$, $y_2 = \min(H, y + 1.25h)$. Captures facial transition boundaries.

3. EfficientNet Compound Scaling: Depth $d = \alpha^\phi$, Width $w = \beta^\phi$, Resolution $r = \gamma^\phi$, s.t. $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$. Balances capacity uniformly.

4. Grad-CAM Neuron Importance Weights: $\alpha_k^c = (1/Z) \sum_i \sum_j (\partial y^c / \partial A_{i,j}^k)$. Global average pooling of gradients over convolutional feature map A^k.

5. Grad-CAM Class Activation Map: $L_{\text{Grad-CAM}}^c = \text{ReLU}(\sum_k \alpha_k^c A^k)$. Applies ReLU to isolate positive evidence supporting manipulation.

6. Video Temporal Top-k Pooling: $S_{\text{video}} = 0.70 \cdot \max(S_t) + 0.30 \cdot (1/N \sum S_t)$. Protects against dilution of transient 2-second deepfakes across 30-second clips.

6. Technology Stack Summary Matrix

Subsystem	Technology Used	Version	Role in DeepGuard System
Frontend UI	React + Vite + Tailwind CSS	React 18.3 / Vite 5.4	Single-Page Application, responsive cards, theme toggle
3D Graphics	Three.js	0.160+	Cybernetic face hero visualizer with viewport-based pause
Camera Studio	HTML5 WebRTC API	getUserMedia	Live camera capture, front/rear flip, hardware track release
Backend API	FastAPI + Uvicorn	FastAPI 0.100+	Async routes, background video worker, single-port hosting
Computer Vision	OpenCV	4.8+	Haar Cascades, NMS, Laplacian blur check, 25% margin padding
Neural Network	PyTorch (EfficientNet-B0)	Torch 2.0+	5.3M parameter compound-scaled CNN, sub-50ms CPU inference
Explainability	Grad-CAM (torch.autograd)	Custom Engine	Feature[8] class activation heatmaps, 5-zone anatomical risk
PDF Reporting	ReportLab	5.0.1	Cryptographic PDF forensic audit certificate generation